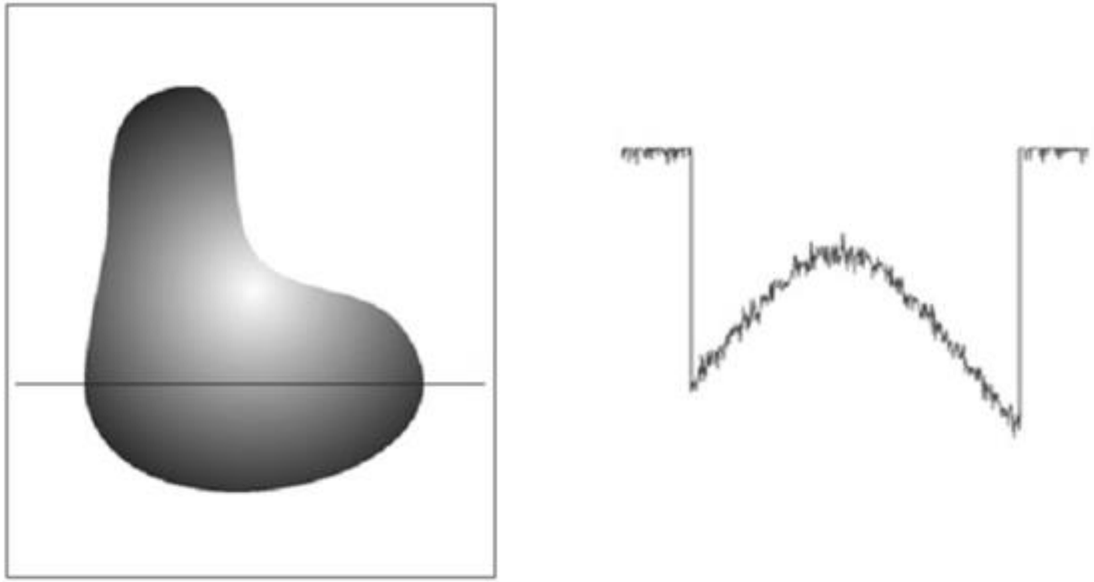
CMPT 732-G200 Practices for Visual Computing

Ali Mahdavi Amiri

Images

- 2D representation.

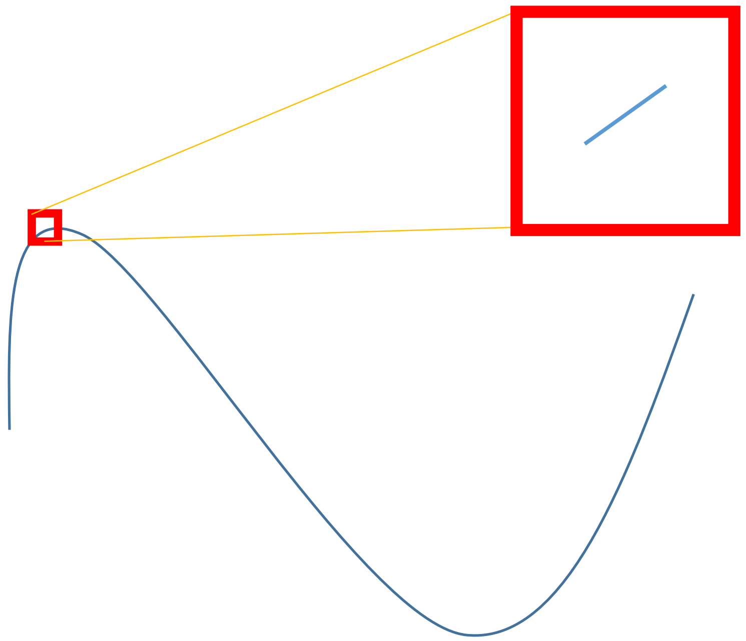


Images

- Each seam is also a curve.

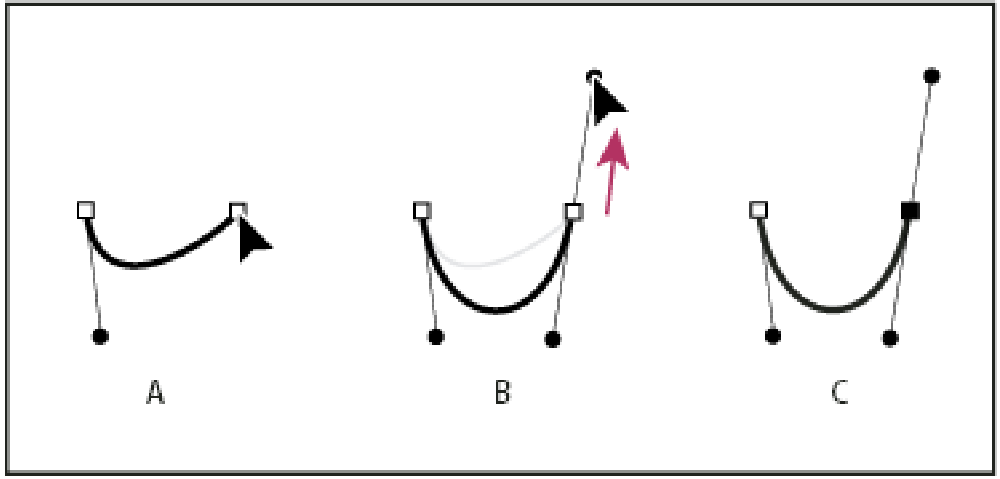
Curves

- A 1D representation that is locally a line but globally has curvature



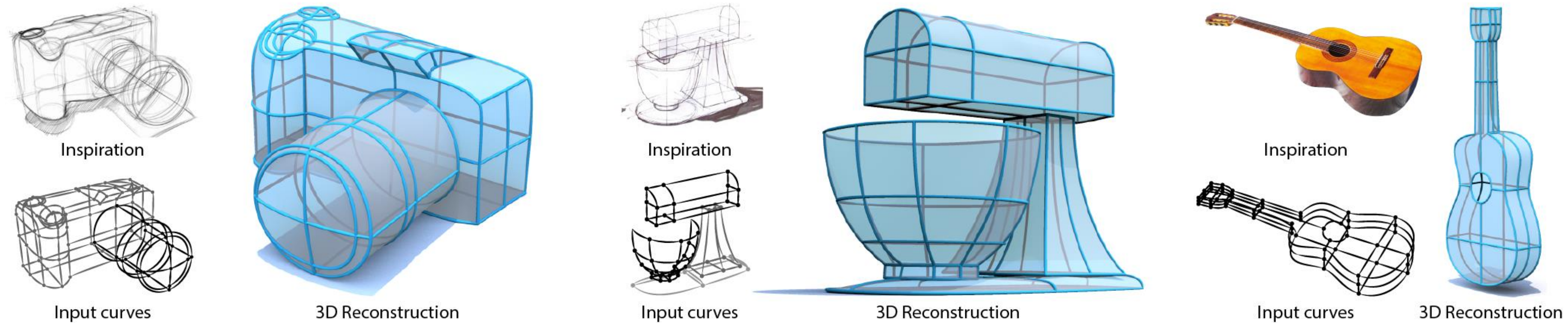
Motivation

- Adobe Illustrator



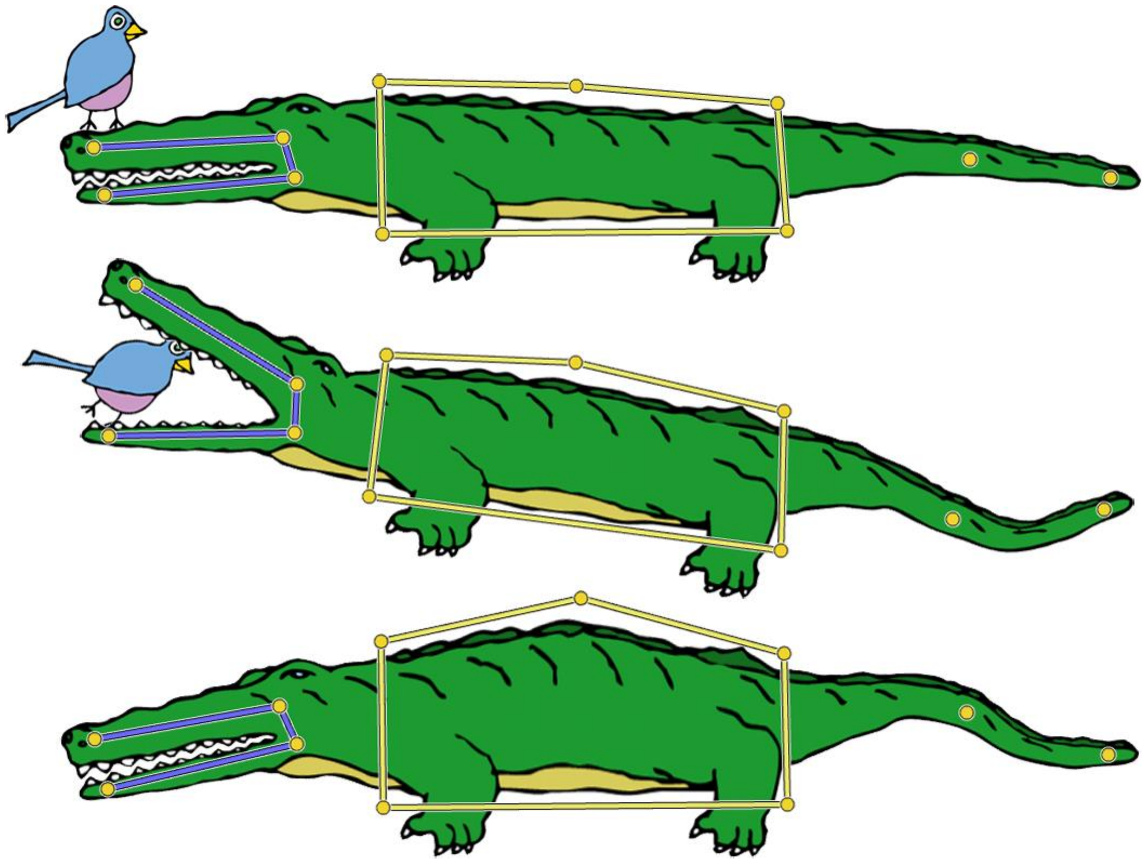
Motivation

- Sketch-Based Modelling



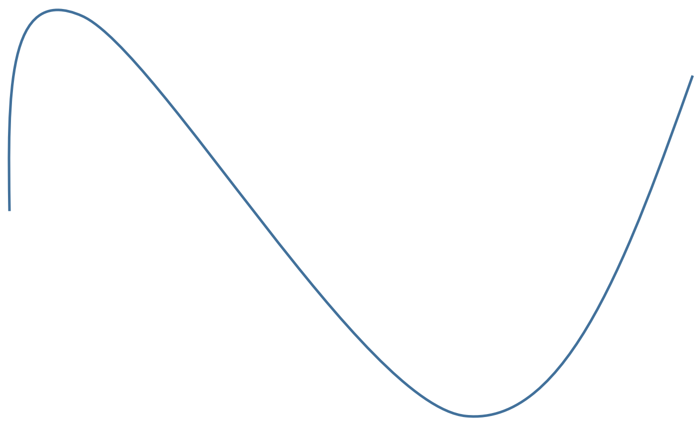
Motivation

- Sketch-based Deformation



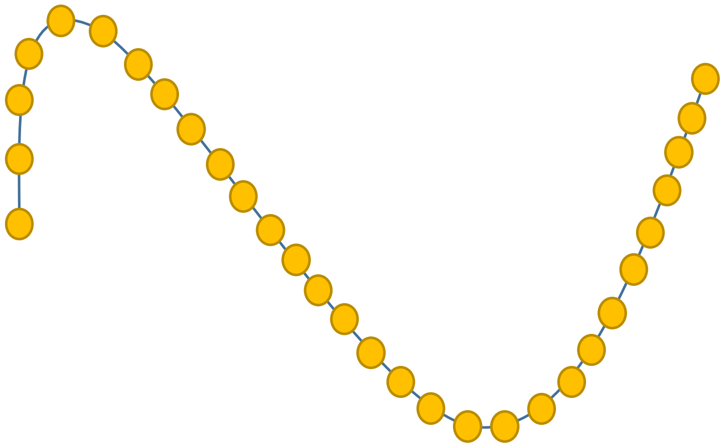
Discrete Curves

- How to represent curves?



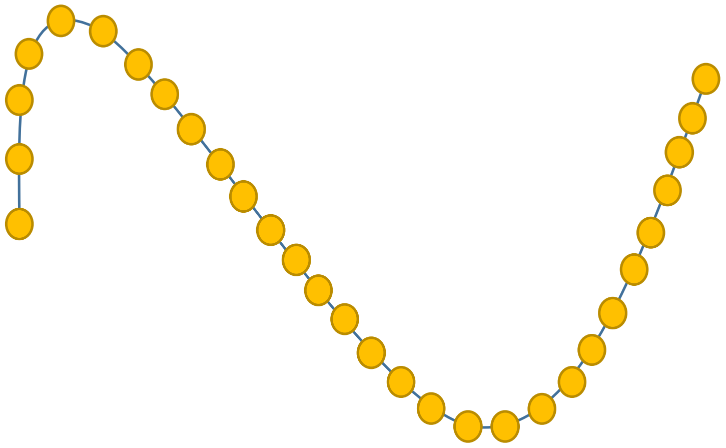
Discrete Curves

- How to represent curves?
  - Discretize the curve with a set of points and connect them by lines
  - Very simple



Discrete Curves

- It is hard to work many control points
  - Hard to manipulate



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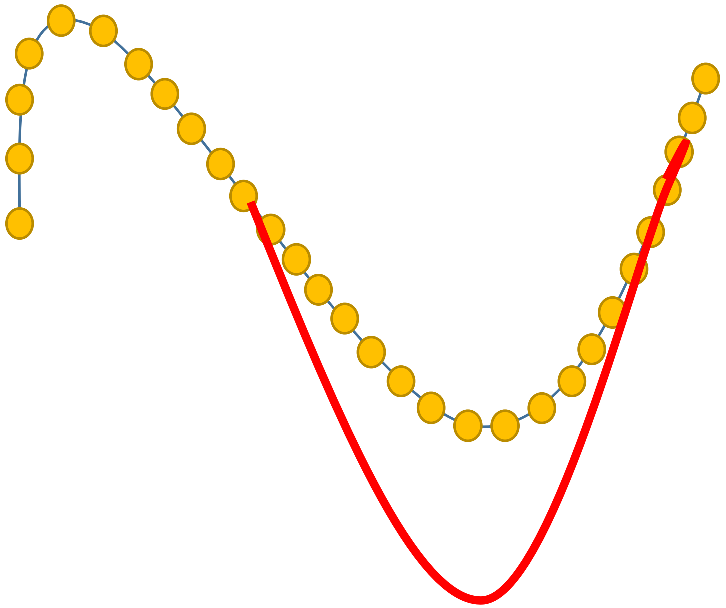

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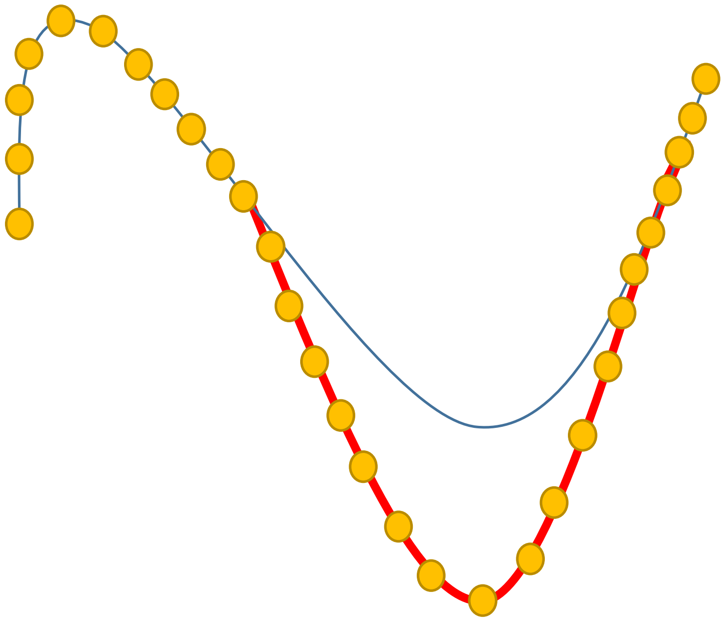
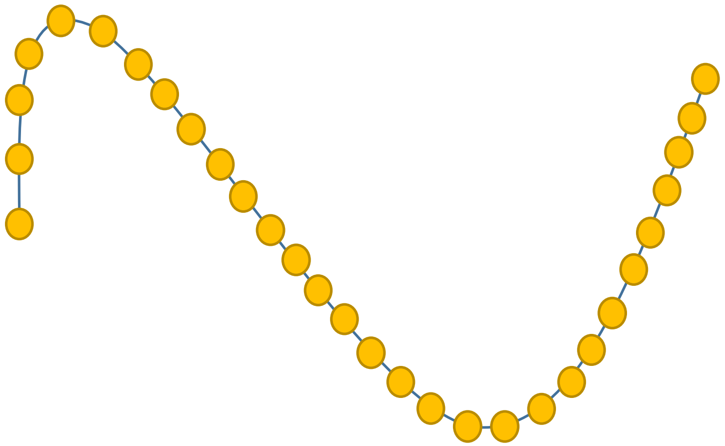


Discrete Curves

- It is hard to work many control points
  - Hard to manipulate
  - Expensive to store/transmit
  - Expensive to address operations (intersections)



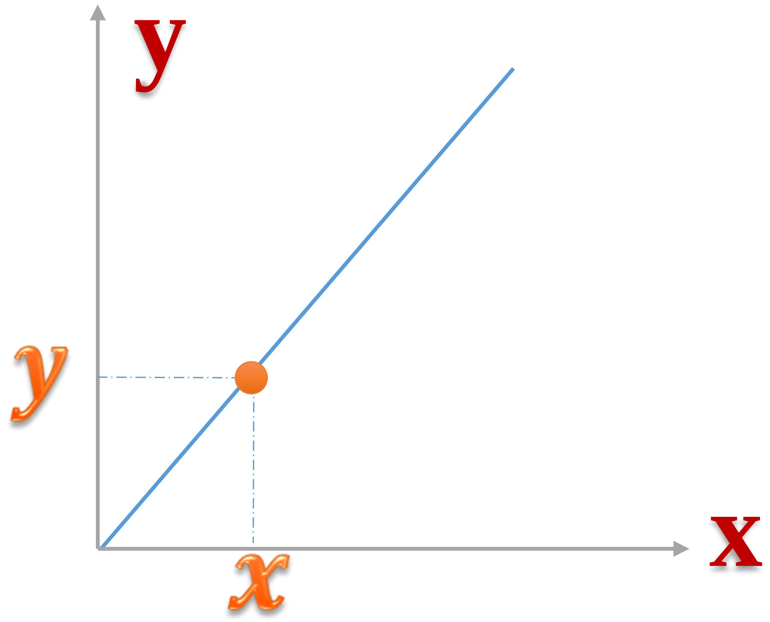
Curves

- What should we do?

- Define a function that represents the curve.

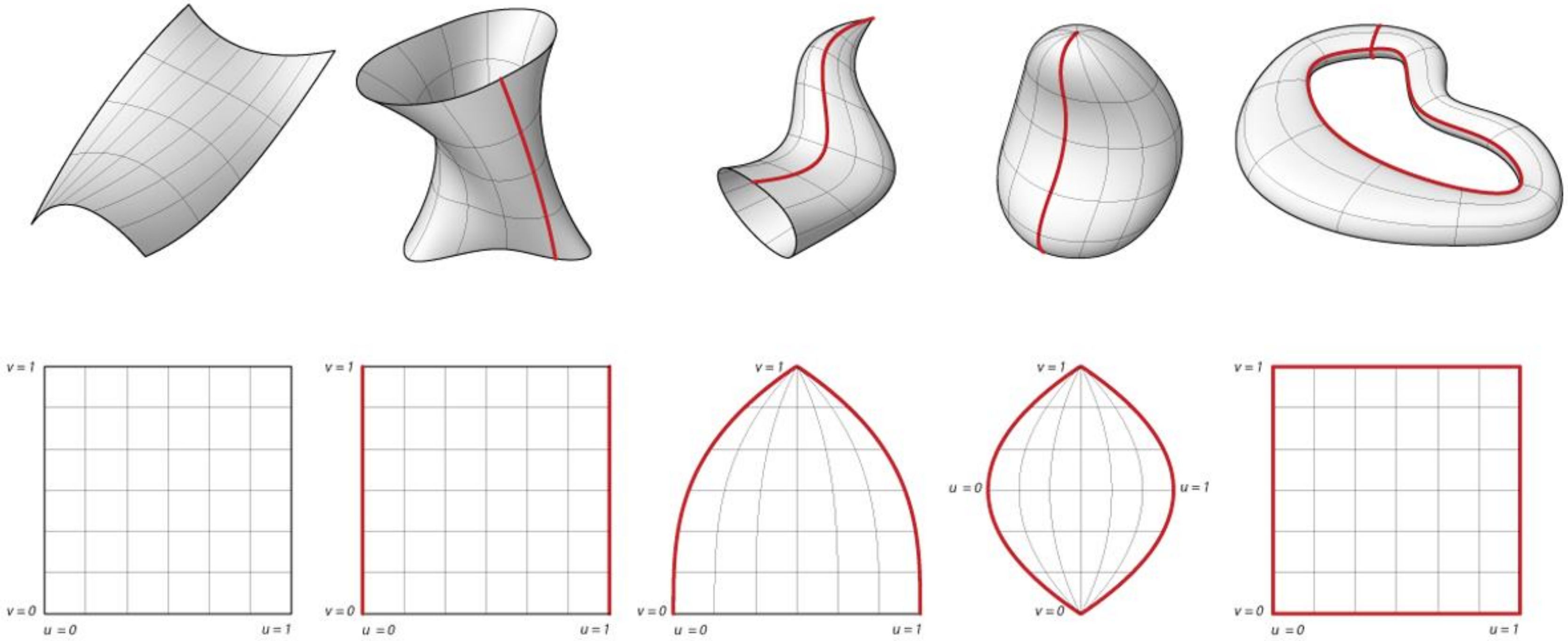
Curves

- Implicit form
- $y = f(x)$, given x find y
- Line: $y = mx + b$



Parametric Models

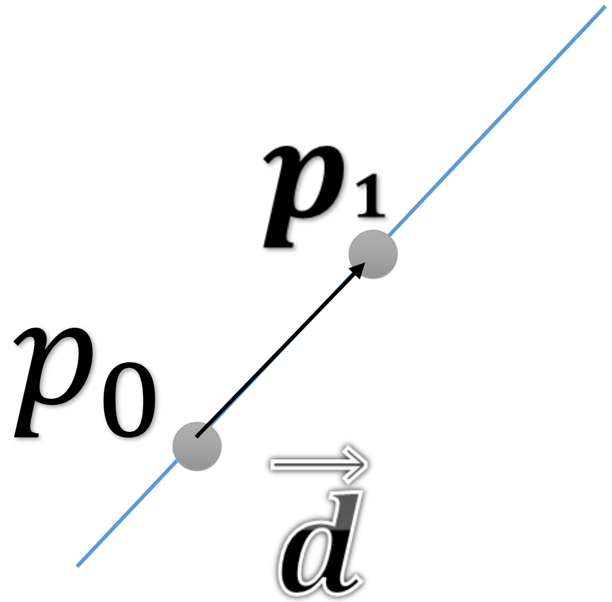
- Splines, Nurbs, B-Splines
- One of the most useful modeling techniques in Graphics and CAD.



Parametric Curves

- $Q(u) = (F_x(u), F_y(u))$

- Example: A parametric Line
  - $P(u) = p_0 + u\vec{d}$
  - $\vec{d} = p_1 - p_0$



Parametric Curves

- $Q(u) = (F_x(u), F_y(u))$

- Example: A parametric Line
  - $P(u) = p_0 + u\vec{d} = ({p_0}_x + (x_1 - x_0)u, {p_0}_y + (y_1 - y_0)u)$
  - $\vec{d} = p_1 - p_0$

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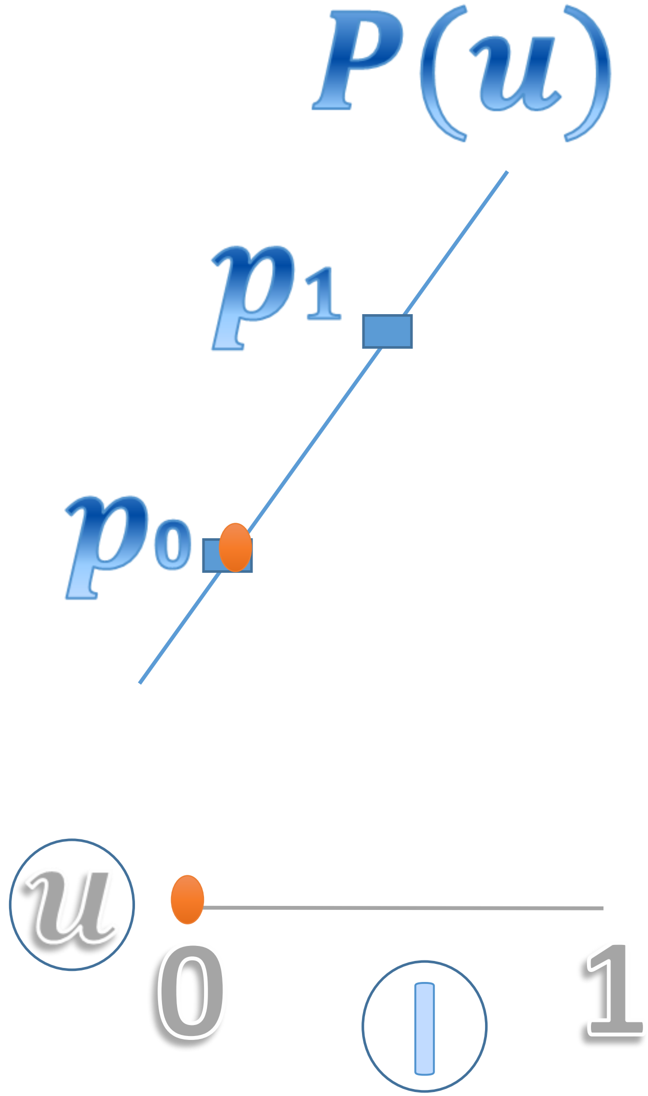
  - $\vec{d} = p_1 - p_0$

Parametric Curves

- $0 \leq u \leq 1$

- $0 \leq u \leq \infty$

- $-\infty \leq u \leq \infty$



Parametric Curves

- Are these parameterizations unique?

Parametric Curves

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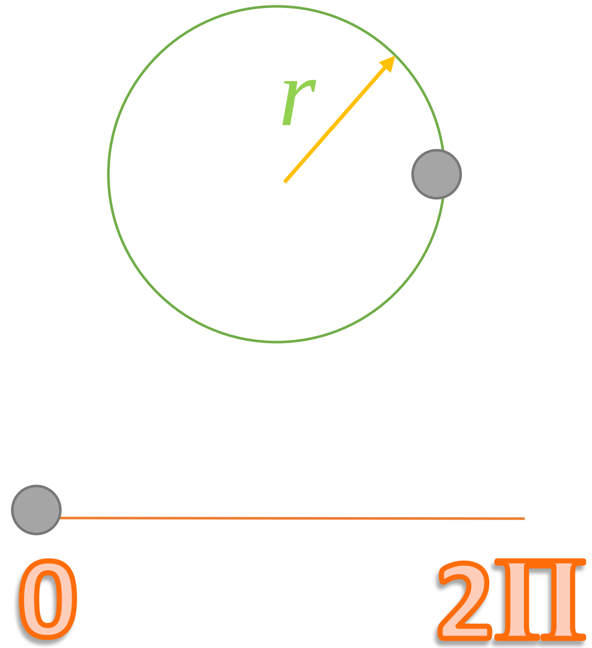
- Circle: $C(u) = \big(rcos(u), rsin(u)\big),$
  $0 \leq u \leq 2\pi$



Parametric Curves

- Are these parameterizations unique?

- Circle: $C(u) = \big(rcos(u), rsin(u)\big)$,
  $$0 \leq u \leq 2\pi$$
- What about $C(u) = \big(rcos(2\pi u), rsin(2\pi u)\big)$,
  $$0 \leq u \leq 1$$

Parametric Curves

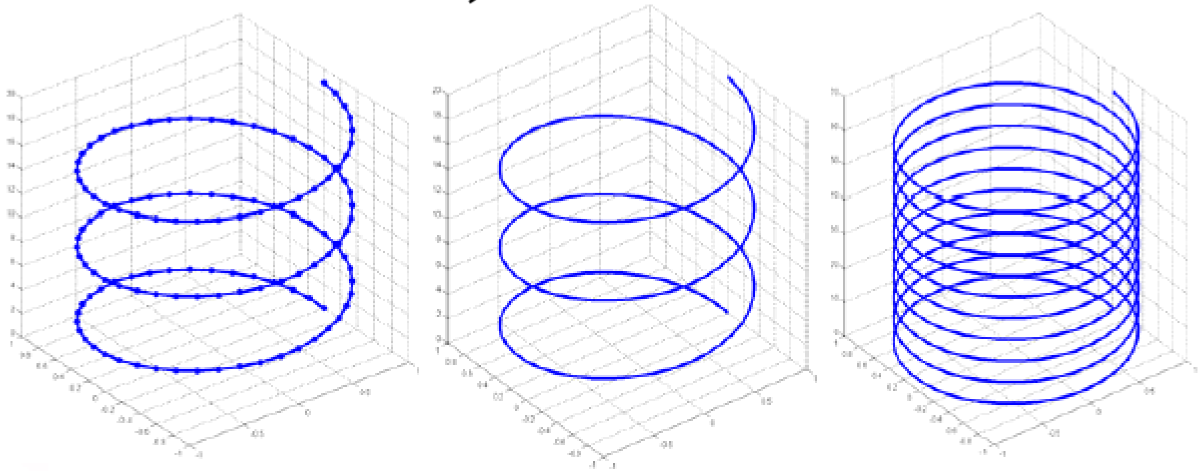
- 3D curves are similar

- $H(u) = \left(F_x(u), F_y(u), F_z(u)\right)$

Parametric Curves

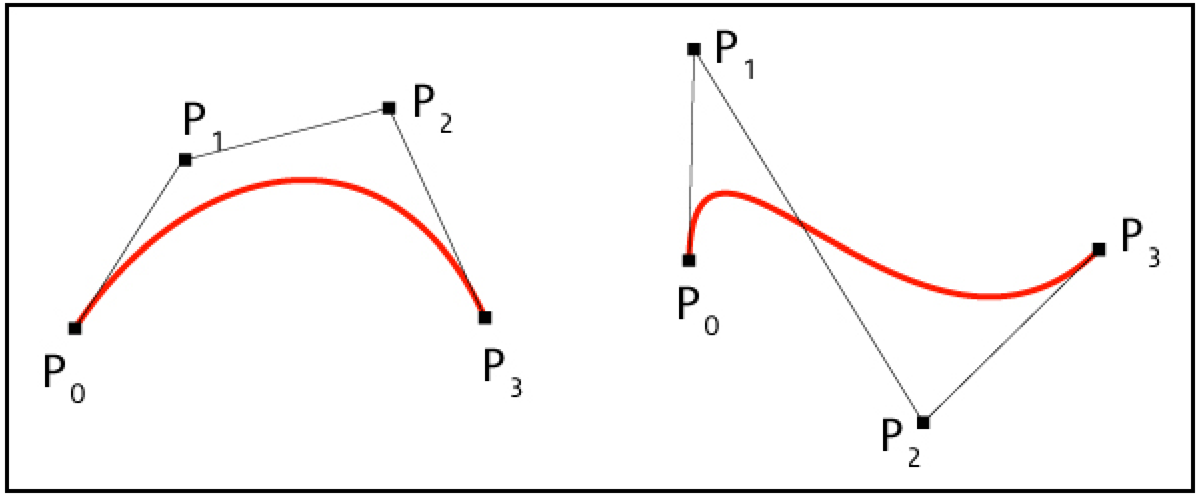
- 3D curves are similar

- $H(u) = \left(F_x(u), F_y(u), F_z(u)\right) = (rcos(u), rsin(u), u)$



Free Form Parametric Curves

- Control points as handles



https://www.jasondavies.com/animated-bezier/

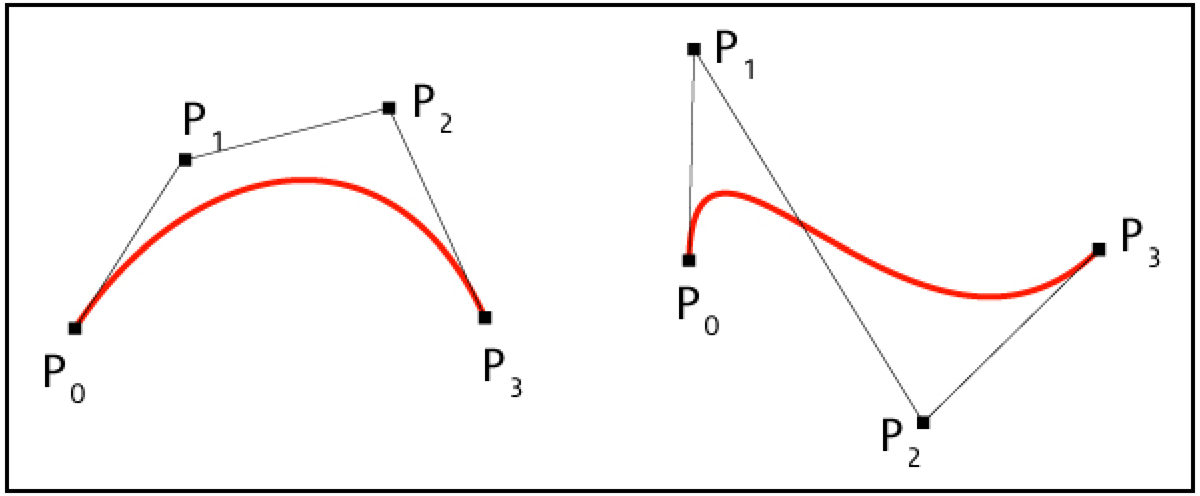
Free Form Parametric Curves

- Demo

Free Form Parametric Curves

- Control points as handles

$$C(u) = \sum_{i=0}^{n} P_i \, B_i(u)$$



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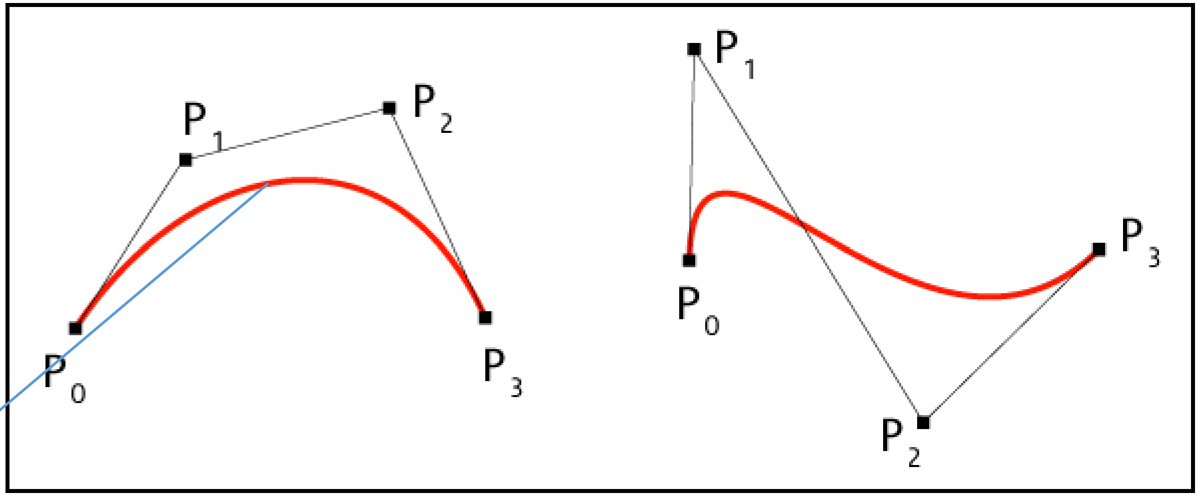

$C(u)$

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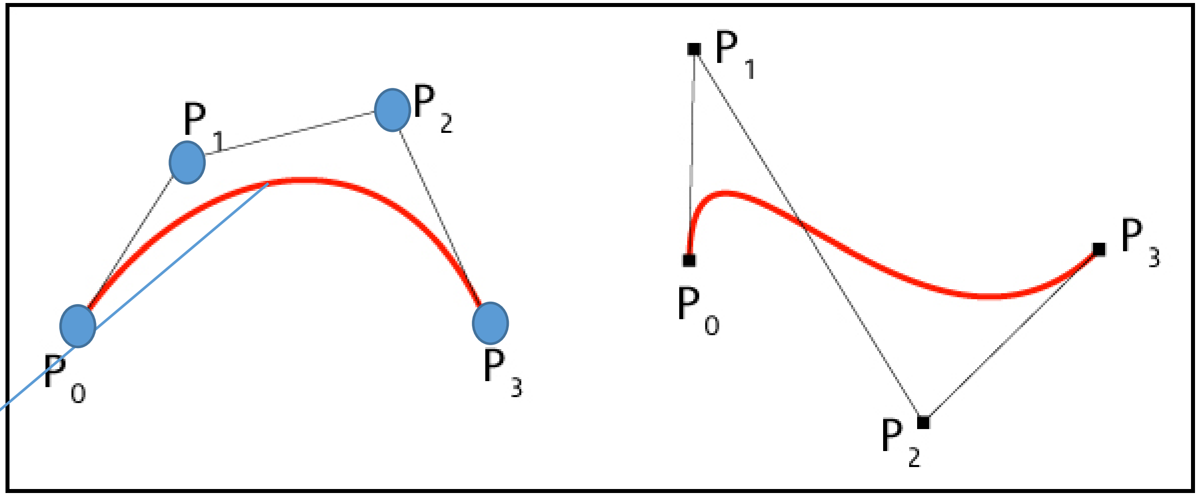

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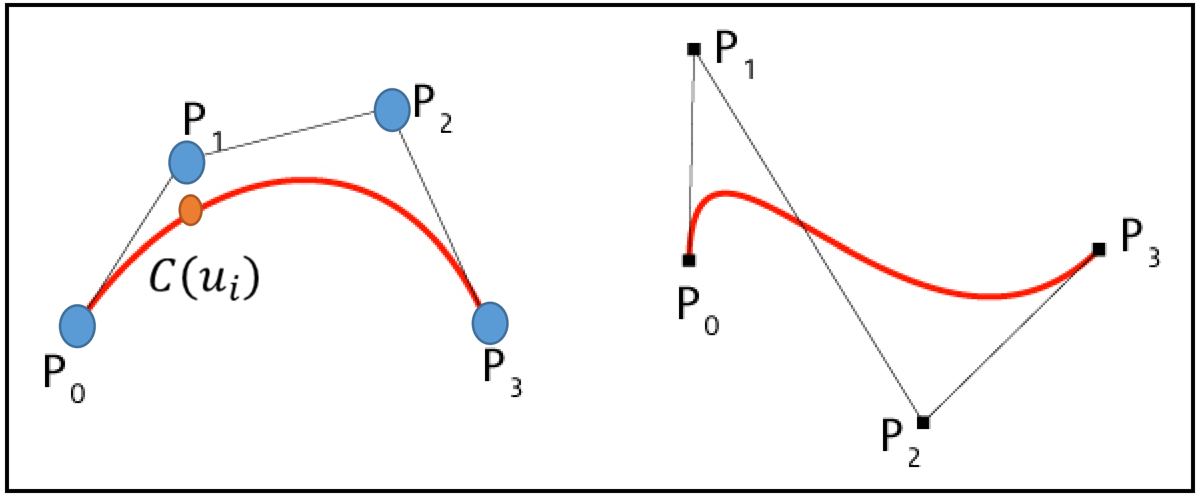

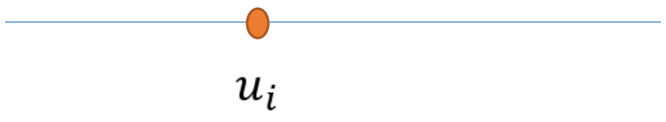

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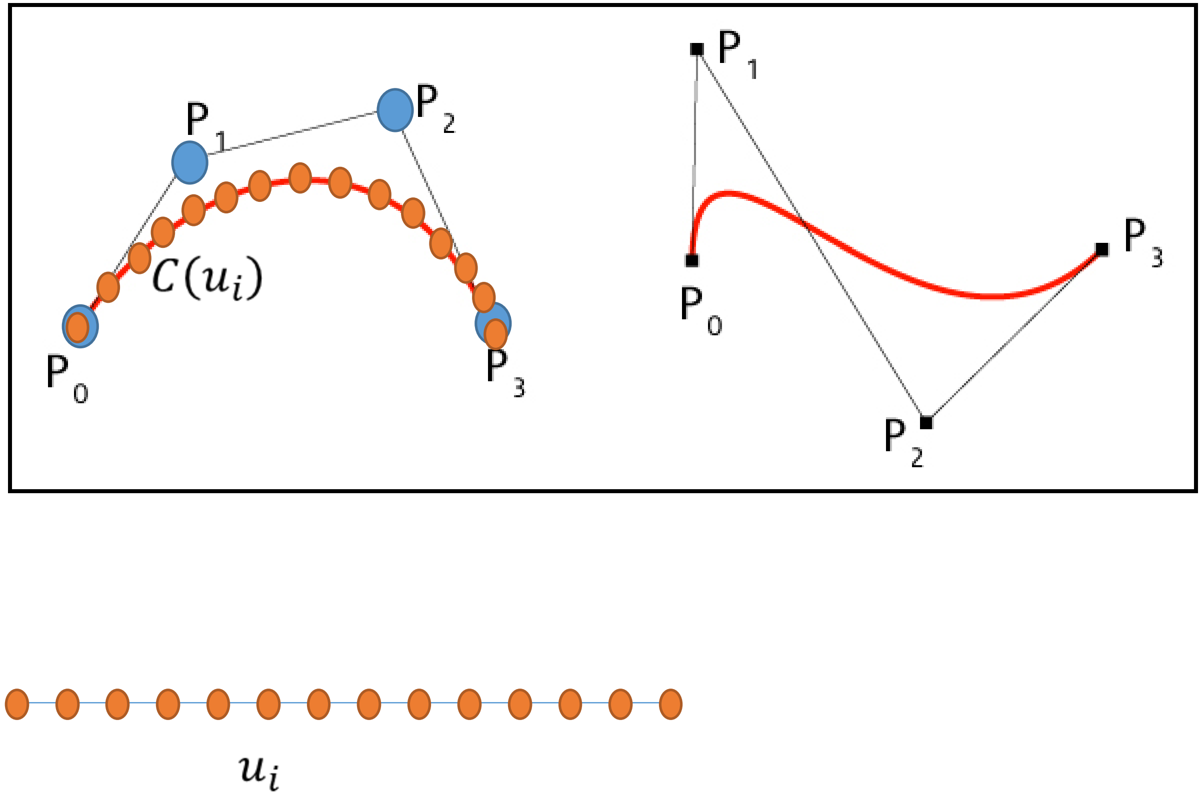

Free Form Parametric Curves

- This means to find a point on the curve all the control points P_i are involved and their effect is controlled by a coefficient determined by $B_i(u)$.

$$C(u) = \sum_{i=0}^{n} P_i \, B_i(u)$$

Free Form Parametric Curves

- How to define basis functions?

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- A summation of points.

Free Form Parametric Curves

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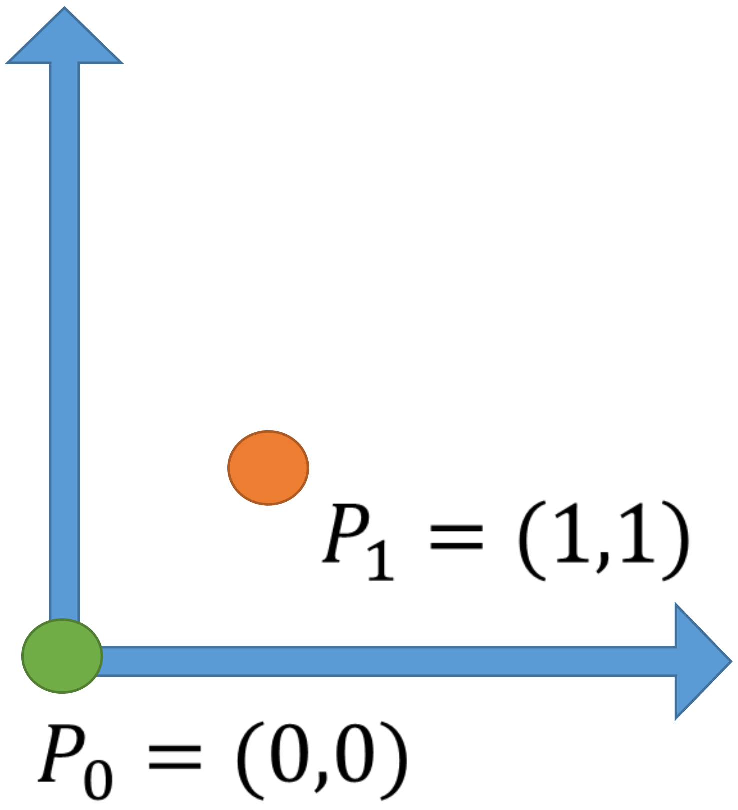
- A summation of points.

- Can we add two points?

Point Summation

- Adding two points?



Point Summation

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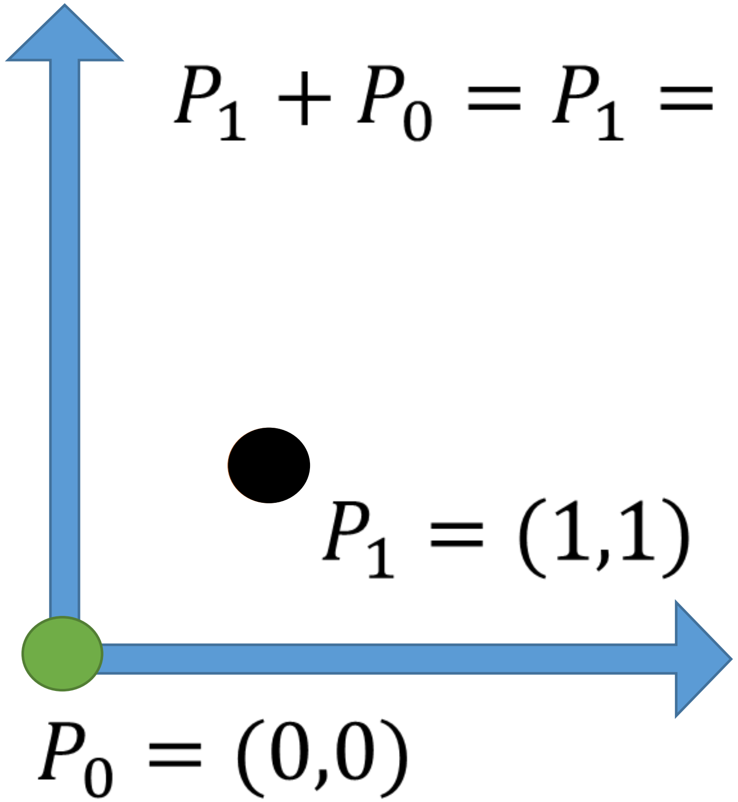
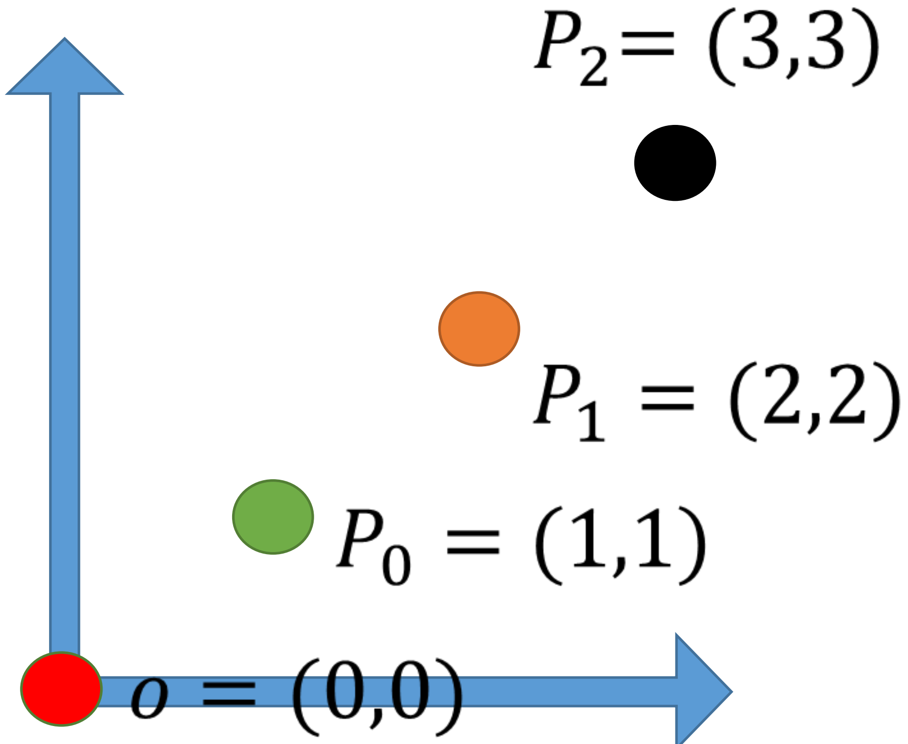


Point Summation

- Adding two points?
  - Change the coordinate system



$$P_1 + P_0 = P_2 = (3,3) \neq P_1$$

Point Summation

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- We cannot add two points directly.

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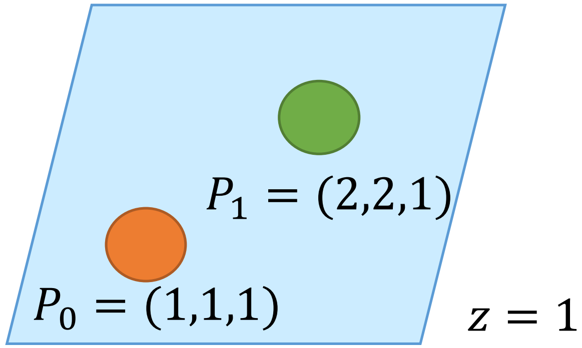

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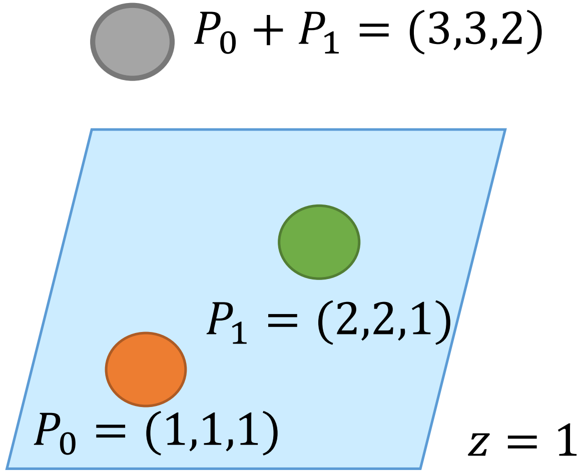

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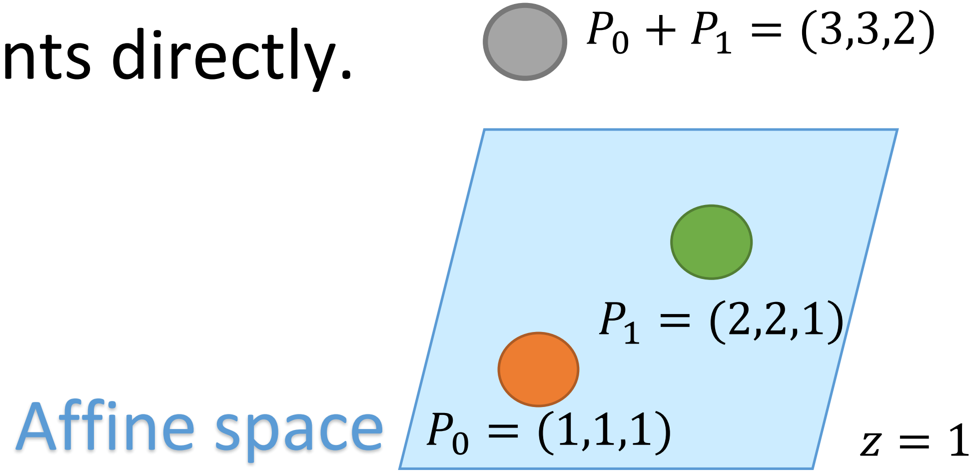

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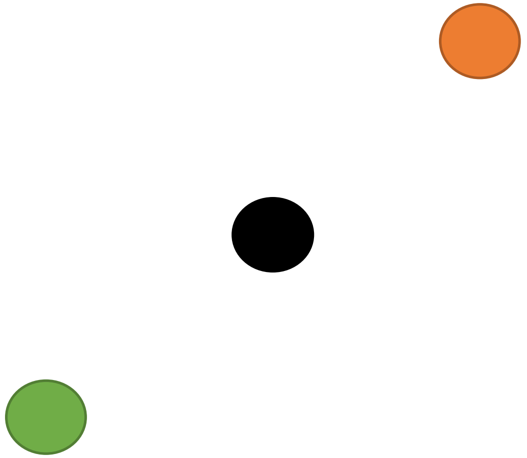

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Why?

Point Summation

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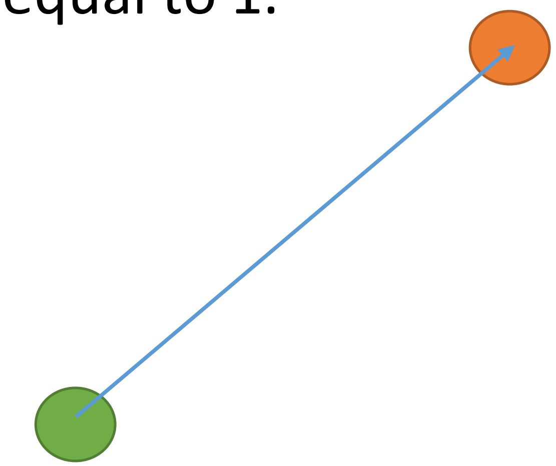

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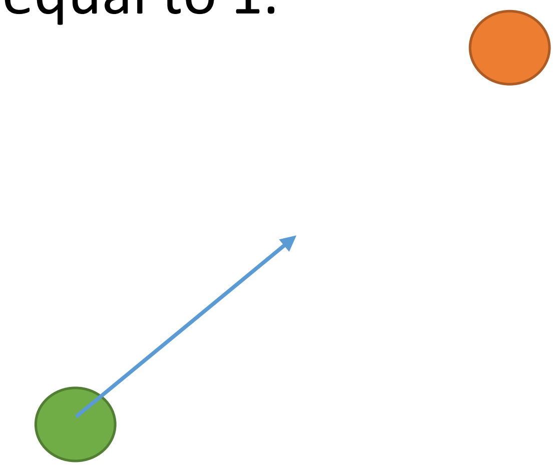

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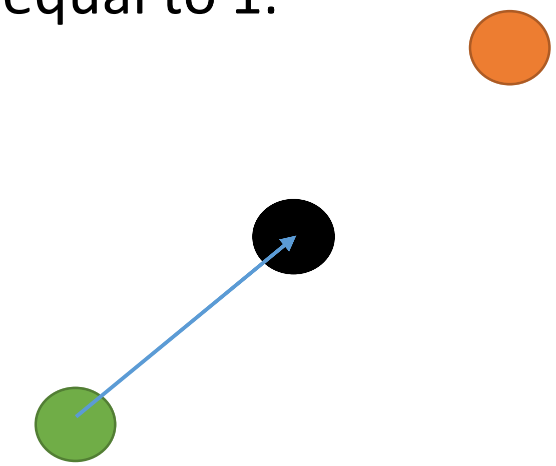

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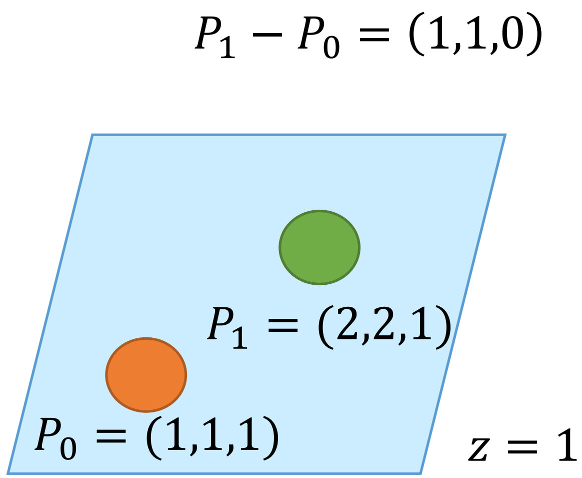

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vector

Point Summation

- Adding several points in general?

$$\alpha_0 P_0 + \alpha_1 P_1 + \cdots + \alpha_n P_n \qquad \alpha_0 + \alpha_1 + \cdots + \alpha_n = 1$$

Point Summation

- Adding several points in general?

$$\alpha_0 P_0 + \alpha_1 P_1 + \cdots + \alpha_n P_n$$

$$\alpha_0 + \alpha_1 + \cdots + \alpha_n = 1$$

$$\alpha_0 = 1 - (\alpha_1 + \cdots + \alpha_n)$$

Point Summation

- Adding several points in general?

$$\alpha_0 P_0 + \alpha_1 P_1 + \cdots + \alpha_n P_n$$

$$\alpha_0 + \alpha_1 + \cdots + \alpha_n = 1$$

$$\alpha_0 = 1 - (\alpha_1 + \cdots + \alpha_n)$$

$$[1 - (\alpha_1 + \cdots + \alpha_n)]P_0 + \alpha_1 P_1 + \cdots + \alpha_n P_n$$

Point Summation

- Adding several points in general?

$$[1-(\alpha_1+\cdots+\alpha_n)]P_0+\alpha_1 P_1+\cdots+\alpha_n P_n =$$

$$P_0+\alpha_1(P_1-P_0)+\cdots+\alpha_n(P_n-P_0)$$

Point Summation

- Adding several points in general?

$$[1-(\alpha_1+\cdots+\alpha_n)]P_0+\alpha_1 P_1+\cdots+\alpha_n P_n=$$

$$P_0+\alpha_1\underbrace{\boxed{(P_1-P_0)}}_{\text{vector}}+\cdots+\alpha_n\underbrace{\boxed{(P_n-P_0)}}_{\text{vector}}$$

Point Summation

- Adding several points in general?

$$[1-(\alpha_1+\cdots+\alpha_n)]P_0+\alpha_1 P_1+\cdots+\alpha_n P_n =$$

$$P_0+\alpha_1\underbrace{\boxed{(P_1-P_0)}}_{v_1}+\cdots+\alpha_n\underbrace{\boxed{(P_n-P_0)}}_{v_n}$$

Affine summation

Point Summation

- Binomial Theorem

$$(x+y)^n = \binom{n}{0} x^n y^0 + \binom{n}{1} x^{n-1} y^1 + \cdots + \binom{n}{n} x^0 y^n$$

Point Summation

- Binomial Theorem

$$(\underbrace{\boxed{x}}_{u} + y)^n = \binom{n}{0} x^n y^0 + \binom{n}{1} x^{n-1} y^1 + \cdots + \binom{n}{n} x^0 y^n$$

Point Summation

- Binomial Theorem

$$(x+\underset{1-u}{\boxed{y}})^n = \binom{n}{0} x^n y^0 + \binom{n}{1} x^{n-1} y^1 + \cdots + \binom{n}{n} x^0 y^n$$

Point Summation

- Binomial Theorem

$$(\boxed{x}+\boxed{y})^n = \binom{n}{0} x^n y^0 + \binom{n}{1} x^{n-1} y^1 + \cdots + \binom{n}{n} x^0 y^n$$

$$u + (1 - u) = 1$$

Point Summation

- Binomial Theorem

$$(x+y)^n = \binom{n}{0} x^n y^0 + \binom{n}{1} x^{n-1} y^1 + \cdots + \binom{n}{n} x^0 y^n$$

$$\boxed{(u+(1-u))^n = \binom{n}{0} u^n (1-u)^0 + \binom{n}{1} u^{n-1} (1-u)^1 + \cdots + \binom{n}{n} u^0 (1-u)^n}$$

Bernstein Polynomial

Point Summation

- Binomial Theorem

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$$(u+(1-u))^n = \binom{n}{0} u^n (1-u)^0 + \binom{n}{1} u^{n-1} (1-u)^1 + \cdots + \binom{n}{n} u^0 (1-u)^n$$

- It can work as a basis function
    - Sum to one
    - Polynomial (easy to calculate)
    - High degree (smooth)

Bezier Curves

- 1972 Pierre Bezier

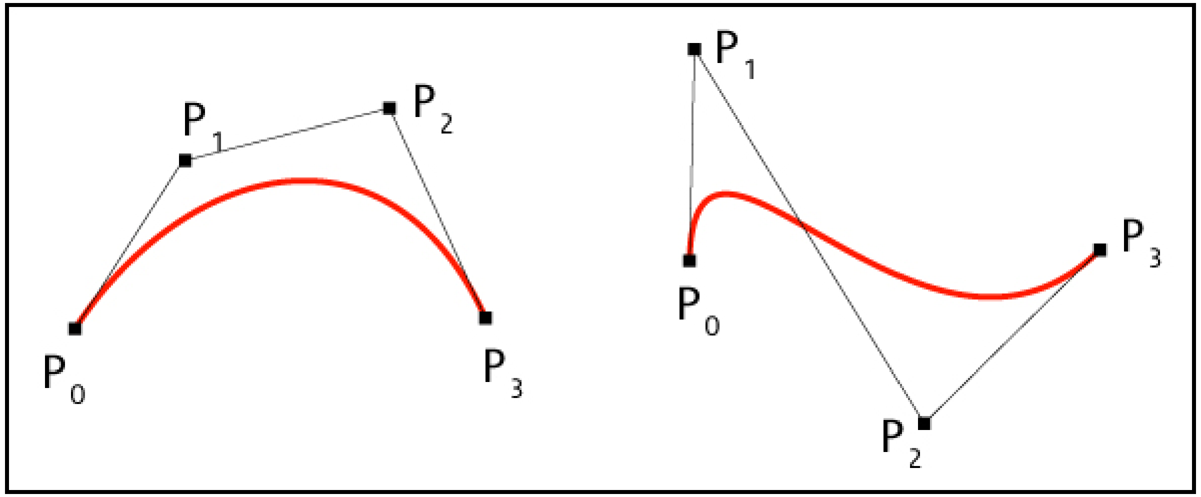
- Renault

- Skin or outer panels of automobiles.

Bezier Curves

$$C(u) = \sum_{i=0}^{n} P_i \, B_{i,n}(u)$$

$$B_{i,n}(u) = \binom{n}{i} (1-u)^{n-i} u^i$$



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Bezier Curves

- Find Bezier Curve with (0,0), (1,1), and (2,0) as control points.

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Bezier Properties

- Bezier Curve must lie in within the convex hull of its control vertices.

$$1 = \binom{d}{0}(1-u)^d + \binom{d}{1}u(1-u)^{d-1} + \cdots + \binom{d}{d-1}u^{d-1}(1-u) + \binom{d}{d}u^d$$

Why?

Bezier Properties

- u and $(1 - u)$ are both nonnegative since $0 \leq u < 1$.
- Coefficients sum to one and nonnegative therefore convex hull property exists.

Bezier Properties

- $d + 1$ control points produce curve of degree d.

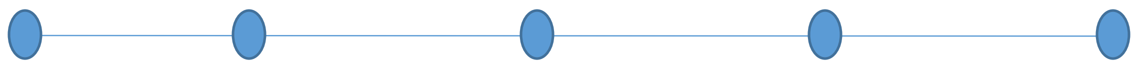
$$C(u) = \sum_{i=0}^{n} P_i\, B_{i,n}(u)$$

$$B_{i,n}(u) = \binom{n}{i} (1-u)^{n-i} u^i$$

i=0 or i=d

Linear precision

- Bezier curve is a line when control points are along a line.



Why?

Avoiding over calculations

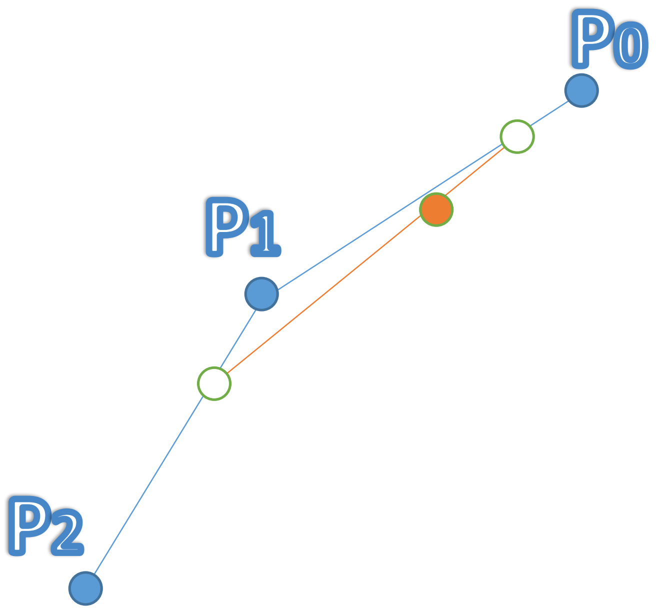
```
for u=0 to 1 step 0.01
   for i=0 to d
      b = Bernstein(i, d, u)
      q = q + b * P[i]
   end
   plot(q)
end
```

$$B_{2,5}(u) = \begin{pmatrix} 5 \\ 2 \end{pmatrix} u^2(1-u)^3 \quad B_{3,5}(u) = \begin{pmatrix} 5 \\ 3 \end{pmatrix} u^3(1-u)^2$$

Over-calculations

de Casteljau algorithm

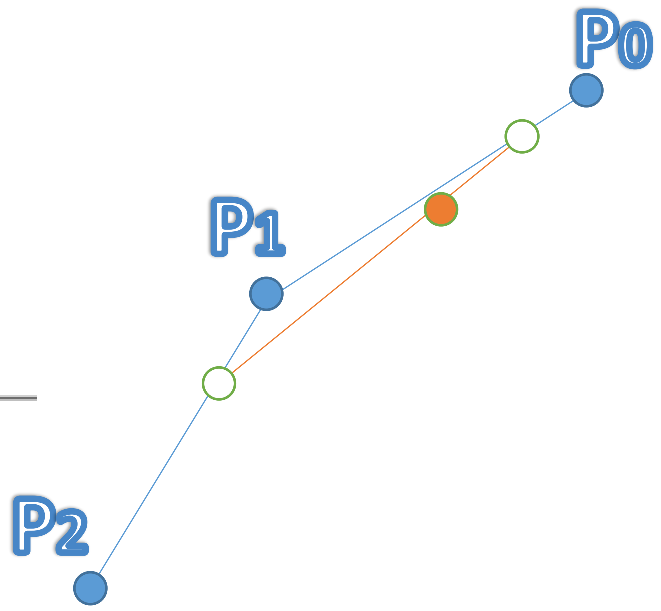
$$
\begin{tikzcd}
P_2 \arrow[dr, "u"] & & \\
P_1 \arrow[r, "1-u"] \arrow[dr, "u"] & P_1^1 \arrow[dr, "u"] & \\
P_0 \arrow[r, "1-u"] & P_0^1 \arrow[r, "1-u"] & P_0^2
\end{tikzcd}
$$



de Casteljau algorithm

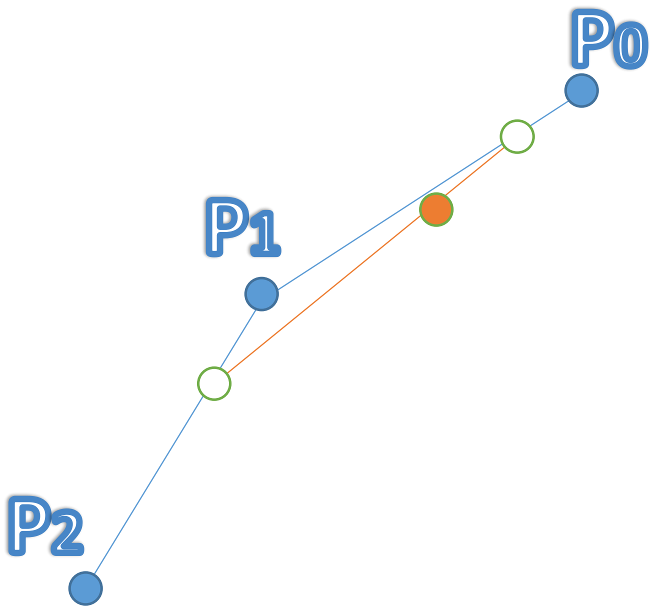
de Casteljau algorithm.

```
// input P[j], d, u
// P[j]:  control point, d:  degree, u:  parameter
//output will be Q(u)
for i = 1 to d
    for j = 0 to d-i
        P[j] = (1-u)*P[j] + u*P[j+1]
    end
end
output p[0]
```



de Casteljau algorithm

$$
\begin{tikzcd}
{\color{red}p_2} & \begin{bmatrix}1\\0\end{bmatrix} \arrow[dr, "\frac{1}{3}"] & & \\
{\color{red}p_1} & \begin{bmatrix}2\\1\end{bmatrix} \arrow[r, "\frac{2}{3}"] \arrow[dr] & \begin{bmatrix}\frac{5}{3}\\\frac{2}{3}\end{bmatrix} \arrow[dr] & \\
{\color{red}p_0} & \begin{bmatrix}3\\5\end{bmatrix} \arrow[r] & \begin{bmatrix}\frac{8}{3}\\\frac{11}{3}\end{bmatrix} \arrow[r] & \begin{bmatrix}\frac{21}{9}\\\frac{24}{9}\end{bmatrix}
\end{tikzcd}
$$



Question

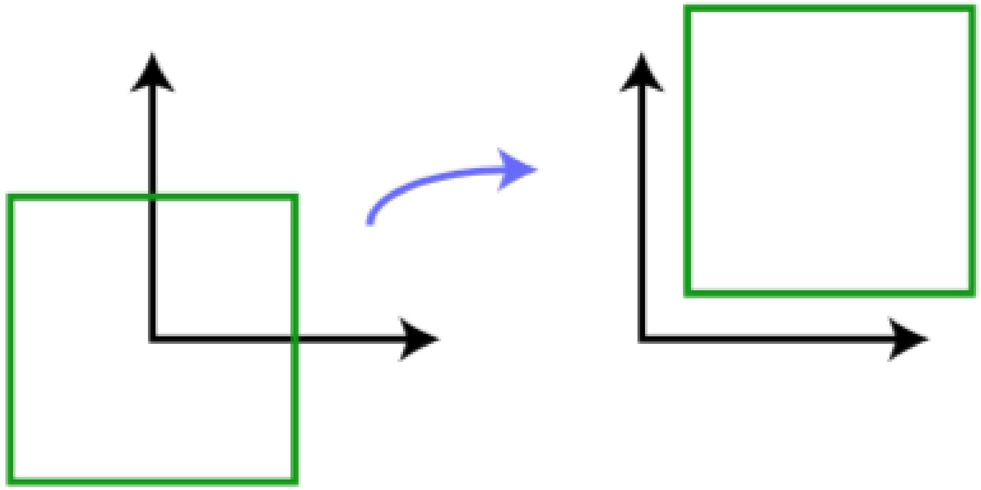
- Prove that a Bezier curve resulting from a transformation is the same as the Bezier curve resulting from the transformation of control points.

Review

- Rigid body transformation: Preserves distances and angles
  - Example: translation, rotation.
- Conformal: Preserves angles
  - Examples: rotation, translation, uniform scaling
- Affine: Preserves parallelism, lines remain lines
  - Examples: translation, rotation, scaling, shear, and reflection

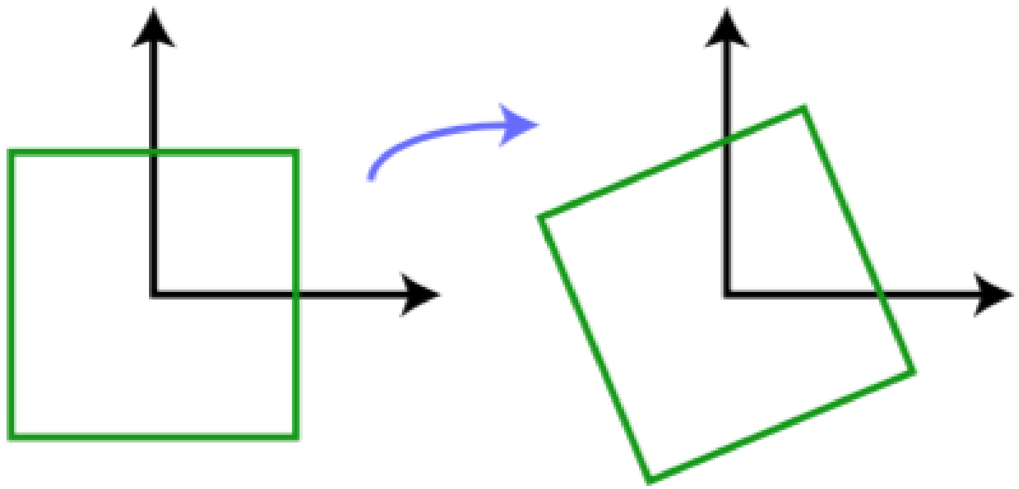
Review

- Translation: $p_1 = p_0 + t$



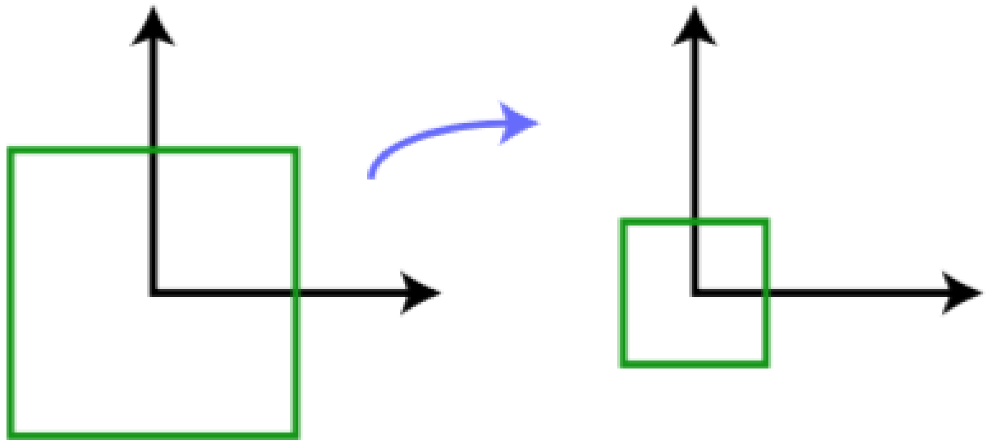
Review

- Rotation clockwise by θ: $p_1 = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} p_0$.



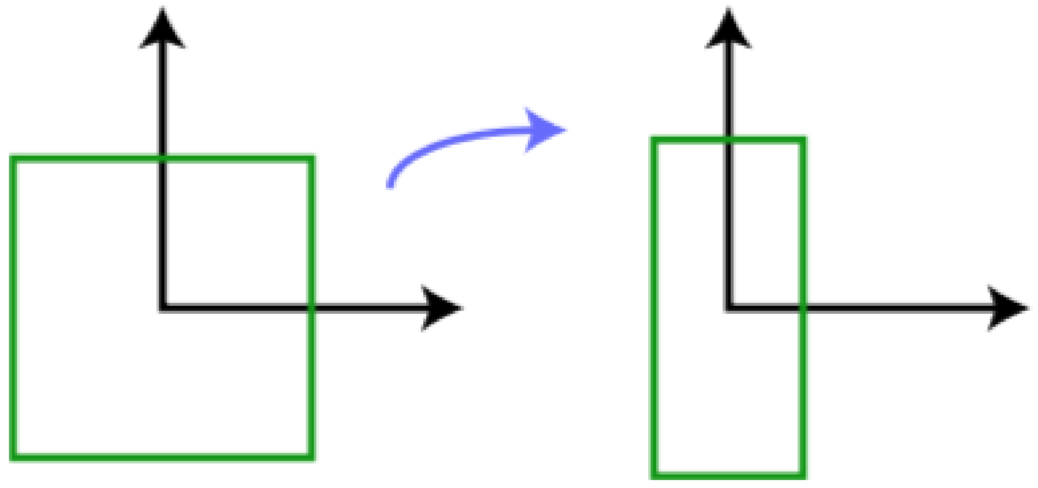
Review

- Uniform scaling by scalar α: $p_1 = \begin{bmatrix} \alpha & 0 \\ 0 & \alpha \end{bmatrix} p_0$.



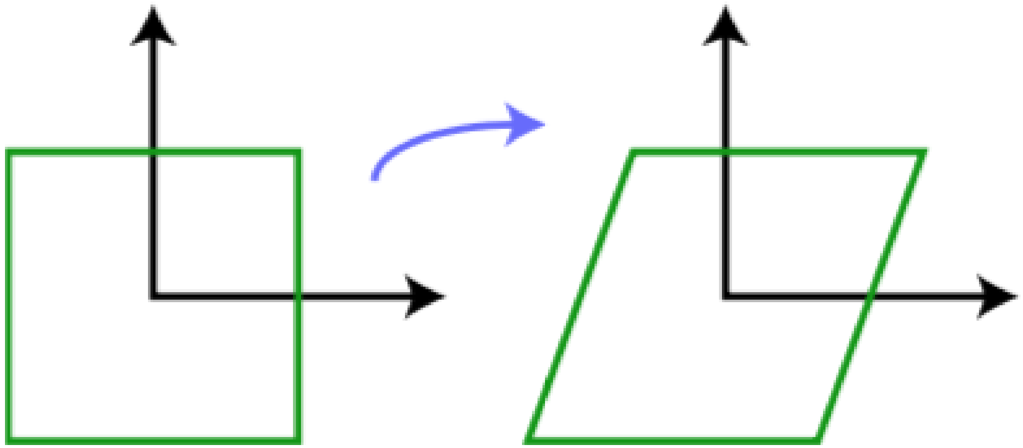
Review

- Non-uniform scaling by scalar α: $p_1 = \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix} p_0$.



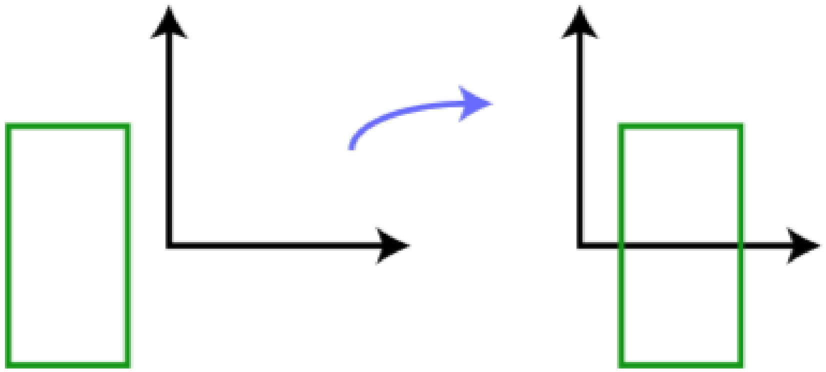
Review

- Shear by scalar h: $p_1 = \begin{bmatrix} 1 & h \\ 0 & 1 \end{bmatrix} p_0$.



Review

- Reflection about the y −axis: $p_1 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} p_0$.



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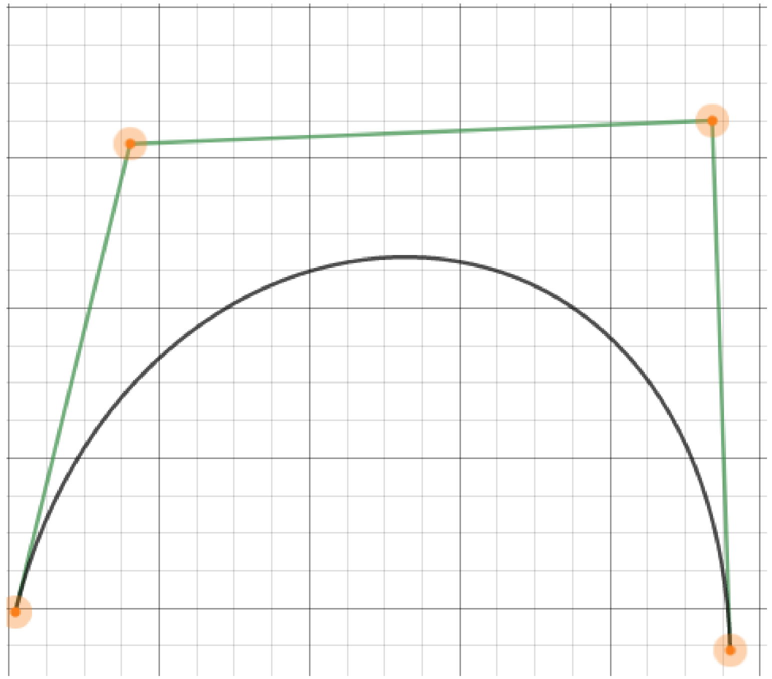
Local VS Global control

- If we move P_1, does the entire curve change or just a part of it?

https://www.desmos.com/calculator/cahqdxeshd



Local VS Global control

- All points are involved in determining the location of a single point on the curve.

$$
\begin{tikzcd}
p_2 \begin{bmatrix}1\\0\end{bmatrix} \arrow[dr, "\frac{1}{3}"] & & \\
p_1 \begin{bmatrix}2\\1\end{bmatrix} \arrow[r, "\frac{2}{3}"] \arrow[dr] & \begin{bmatrix}\frac{5}{3}\\ \frac{2}{3}\end{bmatrix} \arrow[dr] & \\
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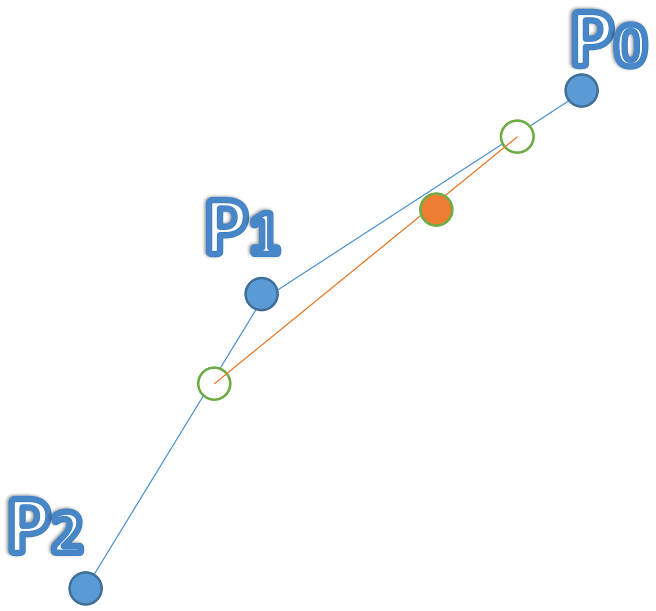
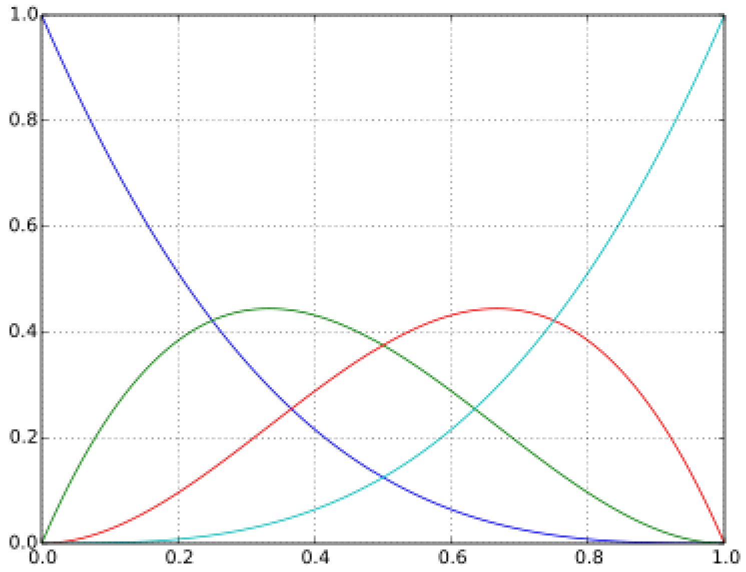
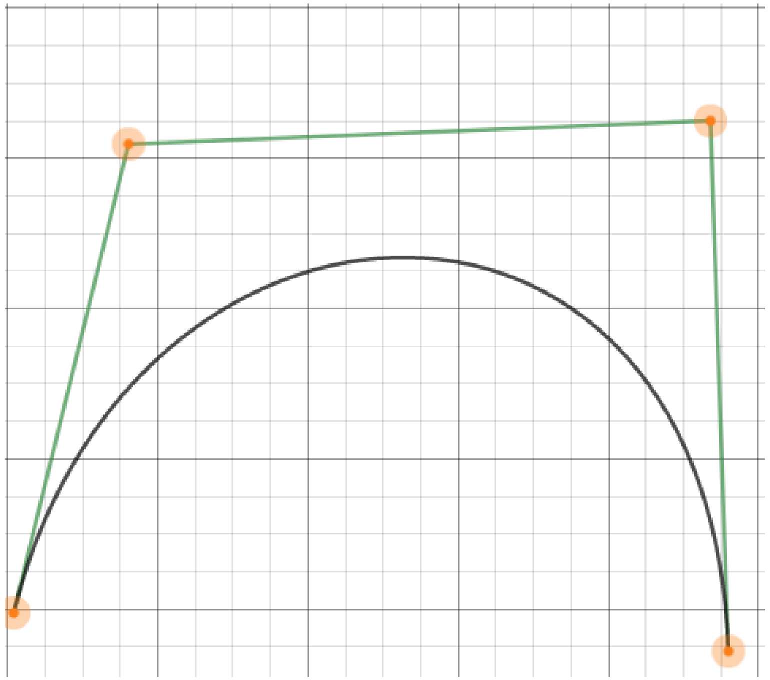


Local VS Global control

- Basis functions are defined on the entire parameter domain.





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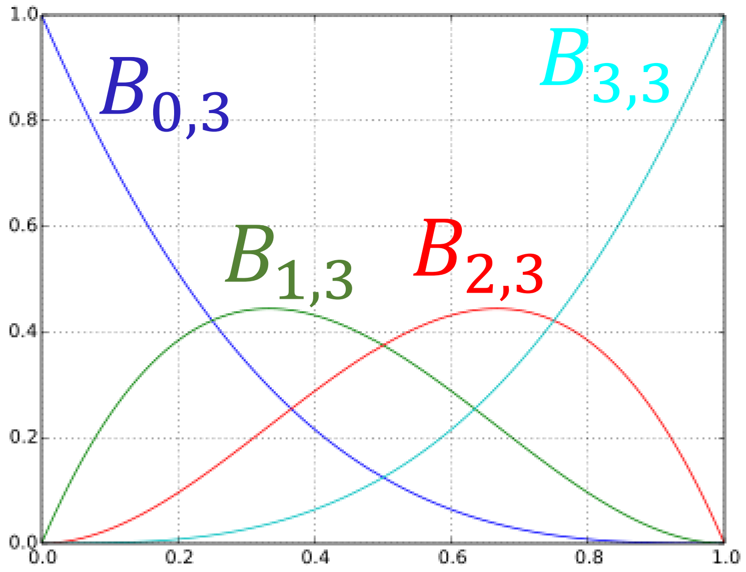
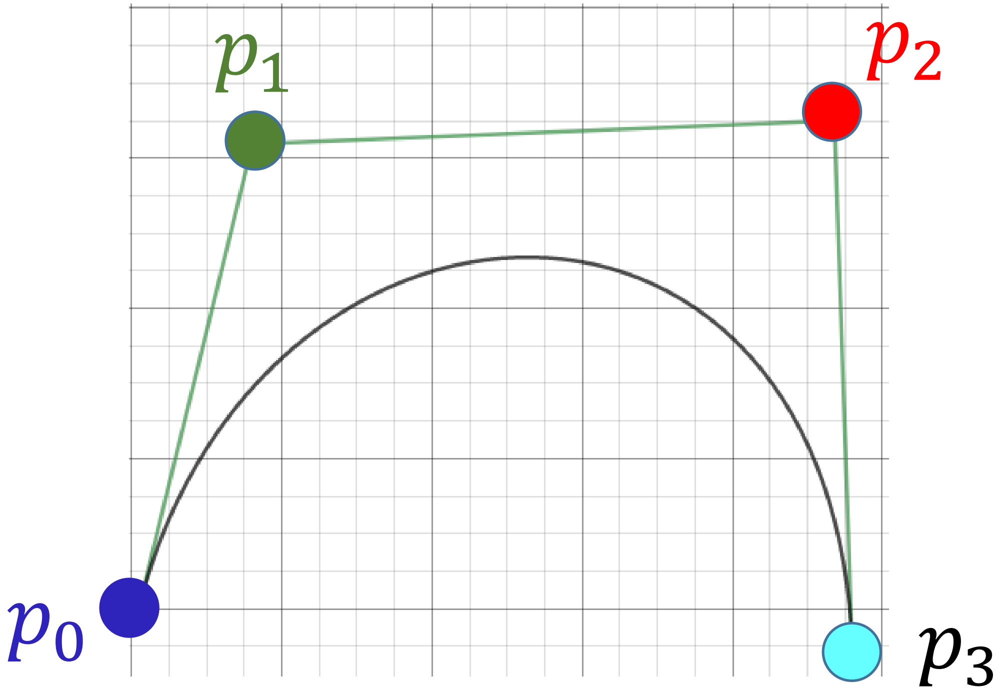
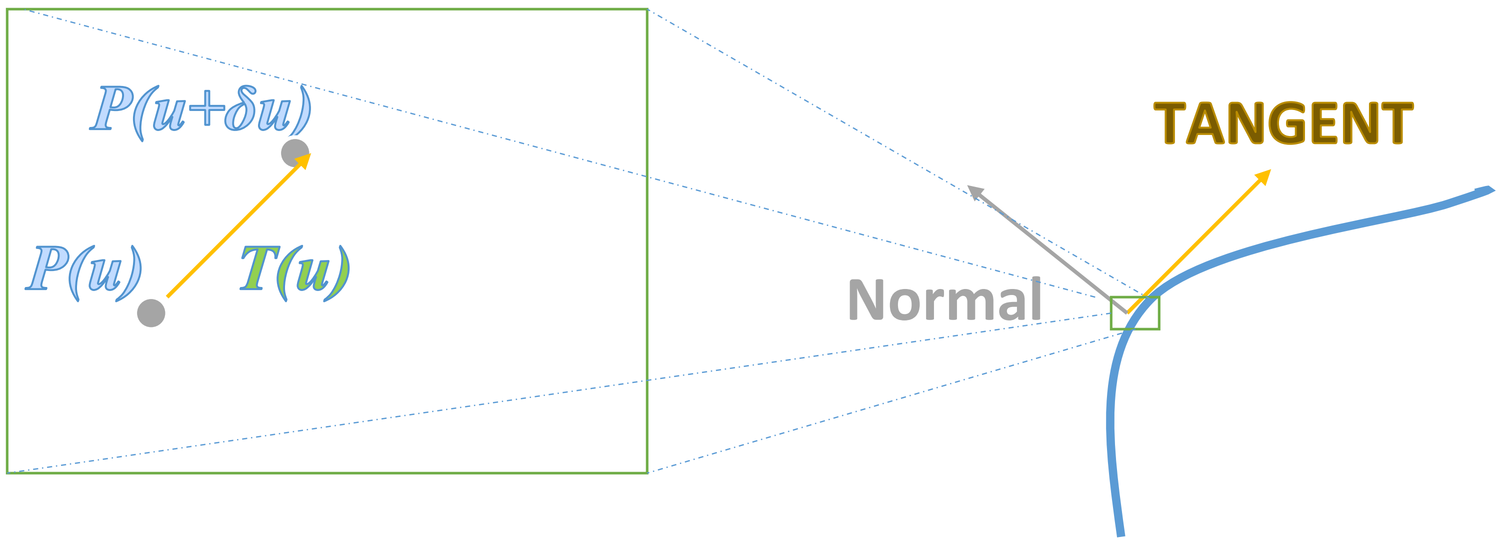




Parametric Curves

- Tangent
  - Find the vector connecting two very close points



Parametric Curves

- Tangent
  - Find the vector connecting two very close points
  - Or find the derivative (they are basically the same thing)

$$T(u) = \lim_{\delta u \to 0}\left(\frac{F_x(u+\delta u) - F_x(u)}{\delta u}, \frac{F_y(u+\delta u) - F_y(u)}{\delta u}\right)$$

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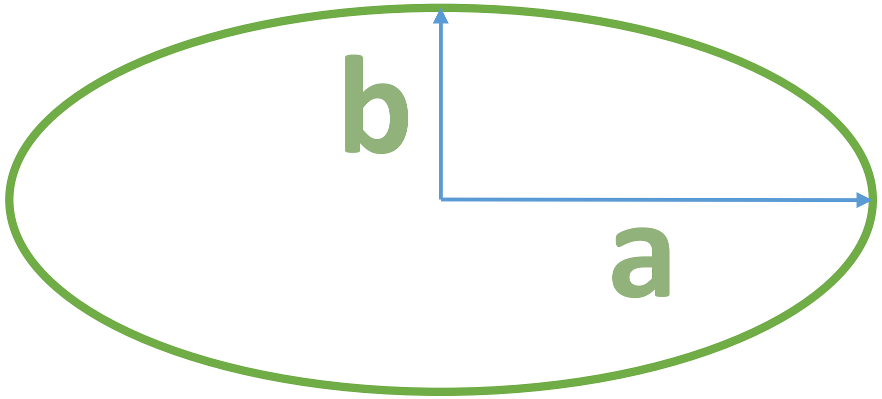
$$T(u) = \frac{dC(u)}{du} = \left(\frac{dC(u)}{du}\right) = \left(\frac{dF_x(u)}{du}, \frac{dF_y(u)}{du}\right) = (\acute{F}_x(u), \acute{F}_y(u))$$

Tangent

- Example: Ellipse

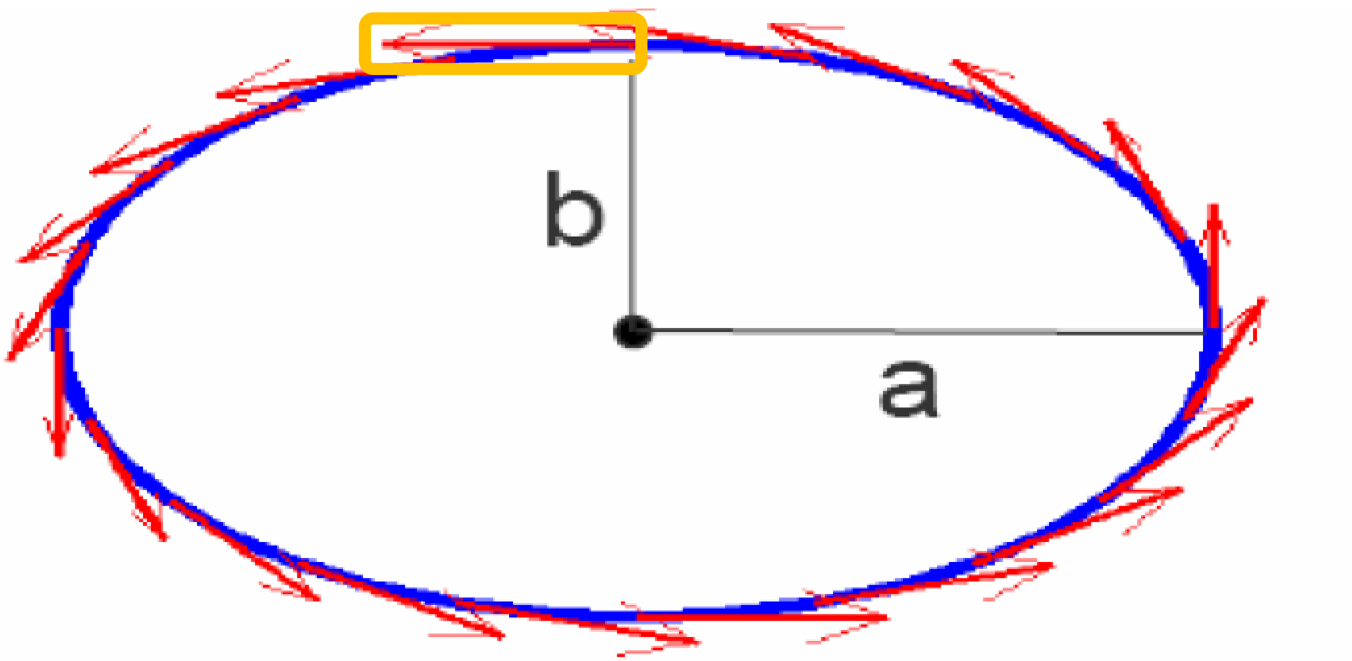


$$Q(u) = (a \cos u, b \sin u) \quad 0 \leq u \leq 2\pi$$

Tangent

$$T(u) = (-a \sin u, b \cos u) \quad 0 \le u \le 2\pi$$

$$T(\pi/2) = (-a, 0)$$

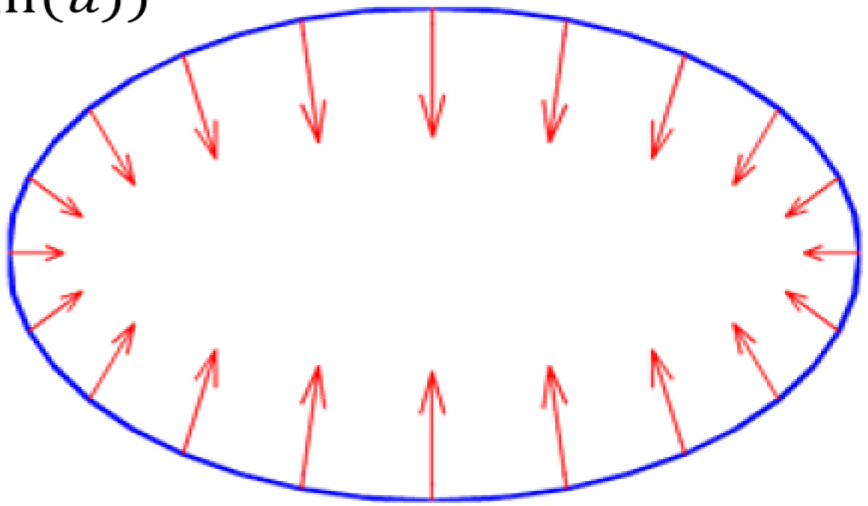


Normal

- Normal vector is Tangent rotated by 90 degrees.

$$N(u) = \begin{bmatrix} \overset{0}{\cos(\frac{\pi}{2})} & \overset{-1}{-\sin(\frac{\pi}{2})} \\ \overset{1}{\sin(\frac{\pi}{2})} & \overset{0}{\cos(\frac{\pi}{2})} \end{bmatrix} \begin{bmatrix} \acute{F}_x(u) \\ \acute{F}_y(u) \end{bmatrix} = \begin{bmatrix} -\acute{F}_y(u) \\ \acute{F}_x(u) \end{bmatrix}$$

$$N(u) = (-b\cos(u) - a\sin(u))$$



Summary

- Discrete Curves
- Parametric Curves
    - Bezier
- Tangent
- Normal